Θ-H Control: Supplementary Material

June 15, 2026

1 Additional Figures

1.1 Θ-H Controller Architecture

Figure S1 illustrates the complete architecture of the Θ-H controller, showing the interaction between the state feedback ($-Kx$), the adaptive Θ action (u_θ), and the BK memory (u_{BK}).

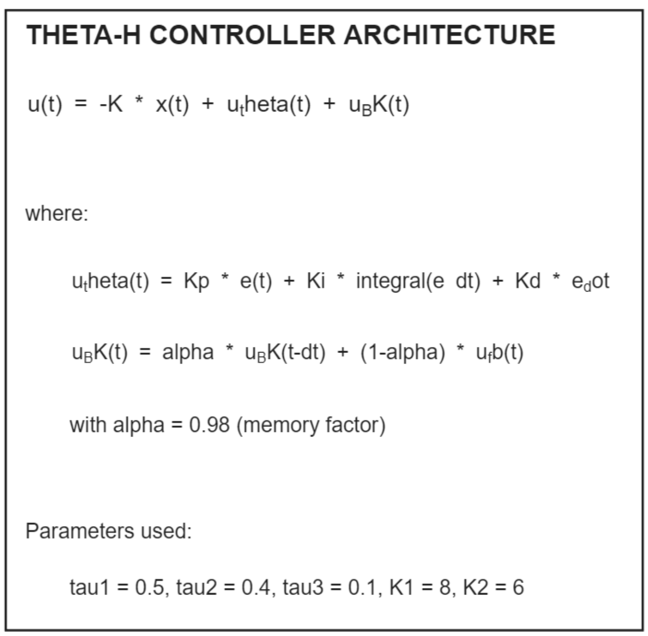

Figure S1: Θ-H controller architecture. The control signal $u(t) = -Kx(t) + u_\theta(t) + u_{BK}(t)$ combines fixed-gain state feedback, adaptive PID action, and low-pass filtered memory. The BK memory ($\alpha = 0.98$) introduces a time-delayed feedback that is essential for the emergent quasi-periodic structure.

1.2 Power Spectrum Comparison

Figure S2 compares the power spectra obtained using different window functions (Hann, Blackman, and Kaiser).

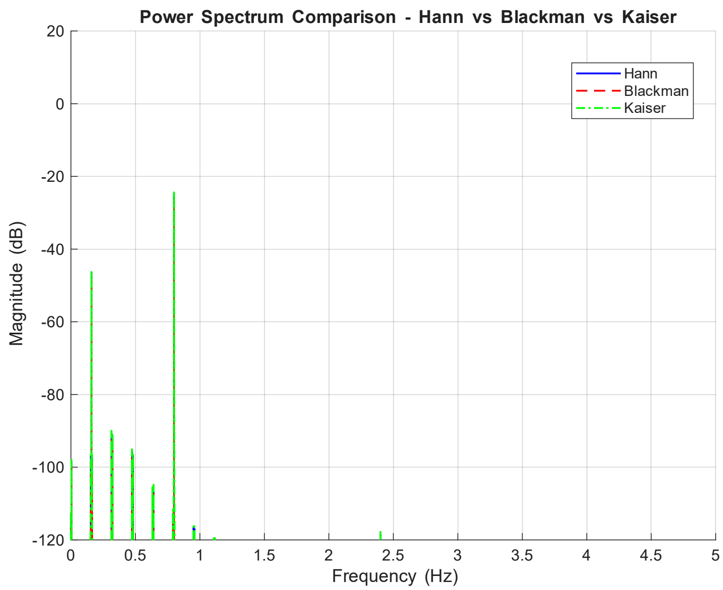

Figure S2: Comparison of power spectra. The re-excitation at high frequencies (2.6-5.0 Hz) persists across all window functions, confirming its physical origin.

1.3 Occupancy Density

Figure S3 shows the occupancy density of the attractor...

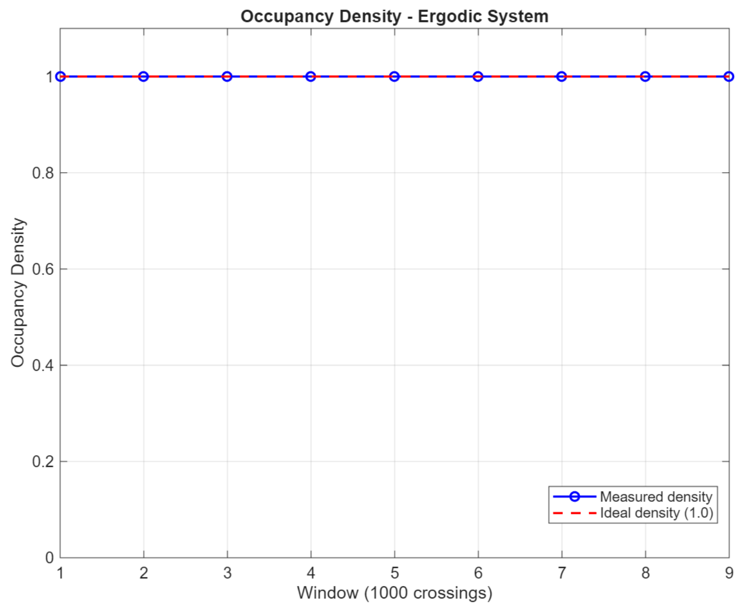

Figure S3: Constant occupancy density (= 1.0) from the first window, indicating an ergodic system.

2 Simulink Model Diagram

Figure S4 shows the complete Simulink model used for the simulations. The model consists of:

- A Step block for the reference signal (setpoint)
- A Sum block for error calculation
- A Mux block to combine the 4 inputs (setpoint, x1, x2, time)
- The Theta-H controller S-Function (4 inputs, 4 outputs)
- The Duffing quartic plant S-Function (2 inputs, 3 outputs)
- A Hit Crossing block for Poincaré section sampling
- Data Type Conversion for signal compatibility
- Scopes for visualization
- To Workspace blocks for data collection

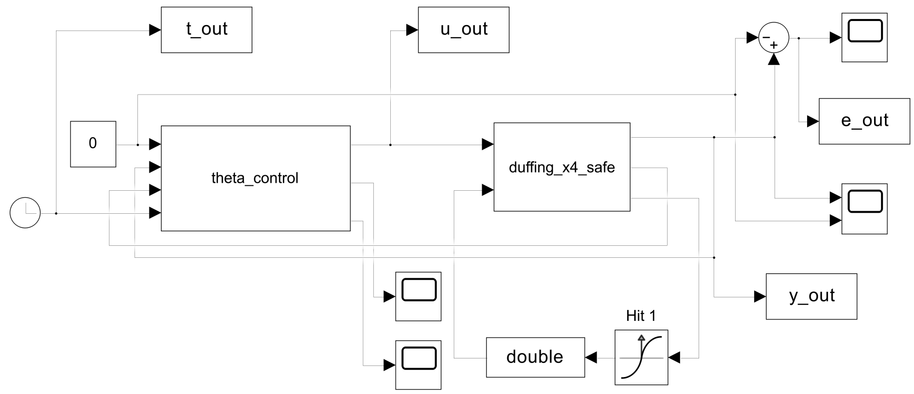

Figure S4: Simulink block diagram of the Θ-H control system. The model implements the complete control loop with Poincaré section sampling.

3 Implementation Details

3.1 Θ-H Parameters

Table 1: Θ-H controller parameters

Parameter	Value
τ_1	0.5
τ_2	0.4
τ_3	0.1
K_1	8
K_2	6
α (BK memory)	0.98

4 Additional Visualizations

4.1 3D Torus Visualization

Figure S5 shows a 3D visualization of the controlled system's trajectory in the phase space (x, v, t). The toroidal structure is clearly visible, confirming the quasi-periodic nature of the attractor. The color gradient represents time evolution.

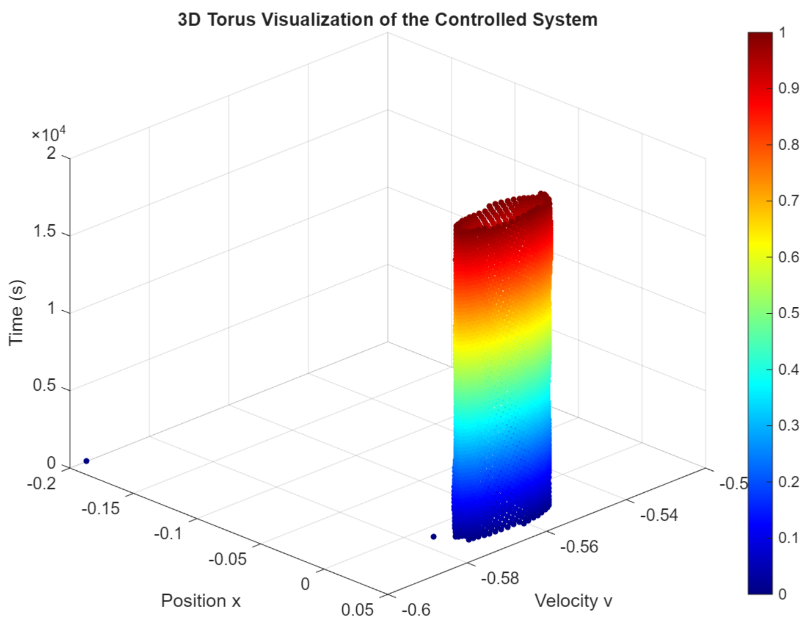

Figure S5: 3D torus visualization of the controlled system. The trajectory evolves on a toroidal surface, characteristic of quasi-periodic dynamics. Color indicates time progression.

4.2 Recurrence Plot

Figure S6 presents the recurrence plot of the controlled system. The regular, non-random structure with parallel diagonal lines is characteristic of quasi-periodic dynamics, distinguishing it from chaotic systems (which exhibit fractal-like patterns).

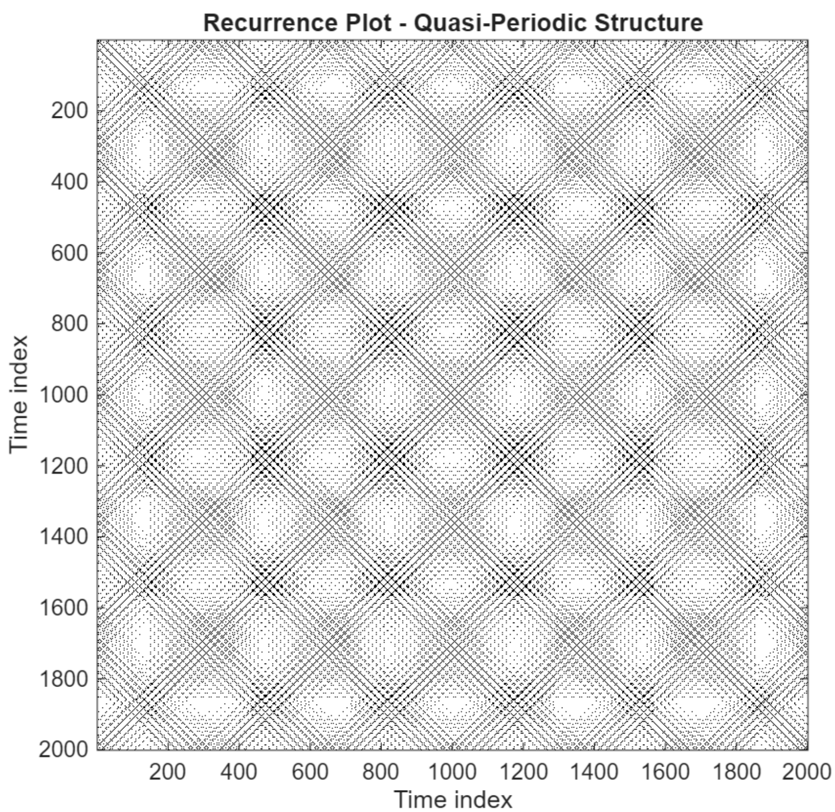

Figure S6: Recurrence plot showing regular, non-random structure with parallel diagonal lines, confirming quasi-periodic behavior.

4.3 Poincaré Section Colored by Phase

Figure S7 shows the Poincaré section colored by phase angle. Two distinct ellipses are clearly visible, intersecting at approximately $x = -0.15$ and $x = 0.1$ (where the velocity $v = 0$). The ellipses have different orientations:

- The larger ellipse has its major axis inclined at approximately 45° relative to the horizontal axis.
- The smaller ellipse has its major axis inclined at approximately 100° (about 55° relative to the first ellipse).

This inclined, intersecting structure indicates that the two quasi-periodic orbits are not independent; they share common points in the Poincaré section, suggesting a possible resonance or bifurcation between the two modes.

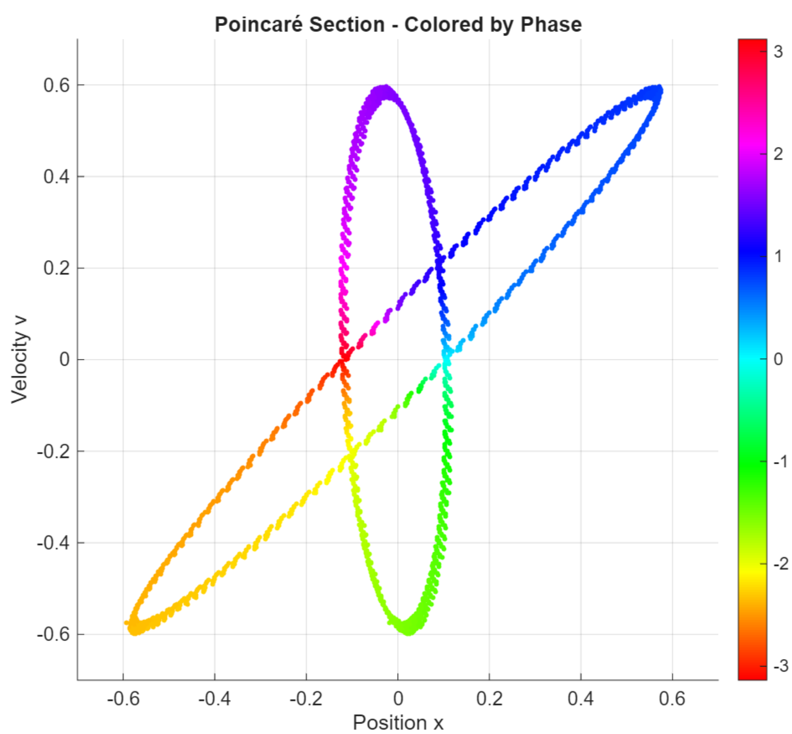

Figure S7: Poincaré section colored by phase angle. Two intersecting ellipses with different orientations are observed, indicating the coexistence of two distinct quasi-periodic orbits.

4.4 Computational Resources

All simulations were performed using MATLAB R2025b with a fixed-step solver (ode4, $dt = 0.001$ s) on a Intel Core i7-2600K, 16 GB RAM, NVIDIA T600, running Windows 11 machine machine.

5 Data Availability

The complete source code and raw data are available at:
https://github.com/marconi-unicamp/theta-h-control